CoffeeScript compiler supports source files where explanation determines the structure of the source code.

[1] Donald Knuth is a computer scientist who is sometimes referred to as the father of the analysis of algorithms. His book *Literate Programming* was published in 1992.

Imagine that Agtron is trying to organize a birthday party for Scruffy. He sends you emails about it, but every time you leave your computer unattended, Scruffy reads your email and finds out what's going on. As a deterrent to Scruffy reading the emails, Agtron wants a simple disguise for all of his emails so that Scruffy will have difficulty reading them. One simple way to disguise messages is by using a cipher called Rot13.

Rot13 is a simple letter-substitution cipher that replaces a letter with the letter 13 letters after it in the alphabet. The built-in string utility for getting character codes can be used. If the character is in the alphabet up to *m*, then add 13 to the character code. If the character is after *m* in the alphabet, then subtract 13 from the character code. Characters can be converted back using the built-in string method. A character is in a specific range regardless of whether it's uppercase or lowercase. Finally, converting a string is done by converting all the characters and joining the results.

With literate CoffeeScript you use the *description* of a program, such as a description of Rot13, as the basis for the program itself. To differentiate them from regular CoffeeScript programs, literate CoffeeScript programs are contained in files with a special extension.

8.1.1. The .litcoffee file extension

Source code contained in a file with a .litcoffee extension is considered a literate CoffeeScript program by the compiler. The key difference between a regular CoffeeScript program and a literate CoffeeScript program is that the comments and the program are reversed. Instead of marking comments with `#`, you leave the comments raw and indent all of the executable code: